

Math 432/532: Introduction to Topology II

HW #5

1. Let $\varphi : V \rightarrow U$ be a linear map. Prove that $V/\text{Ker}(\varphi)$ is isomorphic to $\text{Im}(\varphi)$.
2. Let K be the simplicial complex shown in Figure 6.4 of the book, except with the direction of the arrow on the right flipped, so that $|K|$ is homeomorphic to the Klein bottle. Compute the Betti numbers of K . (You may use the fact that the Klein bottle is connected and non-orientable.)
3. Let Γ be a tree, that is, a connected 1-dimensional simplicial complex with n vertices and $n - 1$ edges for some $n > 0$. Compute the Betti numbers of Γ .
4. Let Γ be the complete graph on 4 vertices, meaning that it is a connected 1-dimensional simplicial complex with 4 vertices and 6 edges. Compute the Betti numbers of Γ .
5. Let Δ_1 be the simplex in $\mathbb{R}^3$ with vertices $\{(0, 0, 0), (1, 0, 0), (0, 1, 0), (0, 0, 1)\}$. Let Δ_2 be the simplex in $\mathbb{R}^3$ with vertices $\{(0, 0, 0), (1, 0, 0), (0, 1, 0), (0, 0, -1)\}$. Let K be the simplicial complex consisting of all proper faces of Δ_1 along with all proper faces of Δ_2 . Thus the realization $|K|$ is homeomorphic to two copies of S^2 , glued together along a copy of D^2 . Compute the Betti numbers of K .